Code No: 117EF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, September - 2021
MACHINE LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. Explain play checkers as a machine learning problem and discuss the design issues.[15]

2.a) Define general boundary and specific boundary of a hypothesis space.
b) Will the candidate elimination algorithm converge to the correct hypothesis? Justify your answer. [7+8]

3. Present the basic ID3 algorithm for learning decision trees and illustrate its operation in detail. [15]

4.a) Illustrate neural network representation used in ALVINN system.
b) What are the differences between standard gradient descent and stochastic gradient descent? [7+8]

5. Explain a procedure to estimate the difference in error between two learning methods with illustration. [15]

6.a) What is the relationship between Bayes theorem and the problem of concept learning?
b) Explain brute force MAP learning algorithm. [8+7]

7. "Nearest neighbor approaches are sensitive to the curse of dimensionality". Support this statement with suitable example. [15]

8. How genetic programming can be used to solve block stacking problem? Explain with an example. [15]

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